



OOAD LAB · IC912B · PROJECT REPORT

IIPS Research Publication & Faculty Performance Information System

Faculty profiles, publications, citations, Ph.D. guidance and a transparent research score for every IIPS faculty member, in one public portal.

42

FACULTY

709

PUBLICATIONS

3,782

CITATIONS

22

PH.D. AWARDED



FACULTY RESEARCH OF
International Institute of Professional Studies
Devi Ahilya Vishwavidyalaya (DAVV), Indore, India

IIPS Research Portal
PUBLICATIONS, CITATIONS AND PH.D. GUIDANCE

HOMEFACULTYRESEARCH SCORESHOW SCORES WORKDATA CHECK

UPDATEData from iips.edu.in, Google Scholar and OpenAlex, checked on 2 Oct 2026.

• Data checked 2 Oct 2026

IIPS Faculty Research Portal

International Institute of Professional Studies, DAVV Indore

42 faculty membersComputer Science and Management

[BROWSE FACULTY](#)[RESEARCH SCORES](#)

PUBLICATIONS: 709CITATIONS: 3,782PATENTS: 2PH.D. AWARDED: 22

LIVE arpanpatra111.github.io/IIPS-Research
CODE github.com/ARPANPATRA111/IIPS-Research
DATA snapshot of 2 Oct 2026

0. At a Glance

One line: a public website that collects every IIPS faculty member's publications from three sources, removes wrong papers, and gives each person a transparent research score out of 100.

What	Where
Live prototype	arpanpatra111.github.io/IIPS-Research
Mirror	iips-research.vercel.app
Source code	github.com/ARPANPATRA111/IIPS-Research
Data snapshot	2 Oct 2026 · 42 faculty · 709 publications · 3,782 citations · 22 Ph.D. awarded
Stack	SvelteKit 2 + Svelte 5, TypeScript, static prerender, Playwright + Vitest, GitHub Pages + Vercel

Lab manual checklist (every task from the manual, in its order)

#	Lab task	Status	Section
1	Problem Statement	Done	§1
2	Stakeholder Analysis	Done	§2
3	Software Requirement Specification (SRS)	Done	§3
4	Functional & Non-functional Requirements	Done	§4, §5
5	10 UML diagrams (use case ... deployment)	Done	§6
6	Database Design (ER + relational schema)	Logical only	§7
7	UI Mockups / Wireframes	Mockups only	§8
8	Layered Architecture	Done	§9
9	Design Patterns	Done	§10
10	Software Design Document (SDD)	Done	§11
11	Prototype · GitHub · Documentation	Done	§12
12	What was not possible, and why	Listed	§13
13	Final Presentation	This document	—

1. Problem Statement

IIPS has 42 faculty across Computer Science and Management, but their research record is **scattered and unverified**:

- The IIPS website lists papers as free text, with no citation counts.
- Google Scholar and OpenAlex have citation data, but **only for some faculty**, and they often attach a **namesake's papers** to the wrong person.
- There is **no single place** to compare research output, and no score whose rules anyone can check.

Goal: build a system that gathers publication data from the IIPS website, Google Scholar and OpenAlex, cleans it, and shows per-faculty profiles, publications, citations, Ph.D. guidance and a **transparent Research Index (0–100)** with rankings. All rules and data matches stay visible.

2. Stakeholder Analysis

Stakeholder	Interest / need	How the system serves them	Influence
Faculty members	Accurate, complete record; fair score	Profile page, evidence for every match, papers set aside with reasons	High
Director / IQAC / admin	NAAC / NBA style research summaries, rankings	Rankings table, department filter, totals on home	High
Students & research scholars	Find a guide by research area	Directory search by name or area, Ph.D. lists	Medium
External visitors / recruiters / collaborators	Quick view of IIPS research strength	Public site, links out to Scholar / OpenAlex / DOI	Low
Maintainer (developer)	Easy refresh, low cost, no server to run	Scripted data pipeline, static hosting, tests	High
Data sources	iips.edu.in, Google Scholar, OpenAlex (external systems)	Read only, cached, rate-limit aware	—

3. Software Requirement Specification (SRS)

Item	Summary
Purpose	One public, read-only portal for IIPS faculty research, with a score whose rules are open.
Scope	In: profiles, publications, citations, h / i10 index, Ph.D. guidance, Research Index, rankings, data-match review. Out: logins, data entry forms, official UGC API / PBAS scoring.
Users	Visitor (student, faculty, admin, public): read only. Maintainer: runs the data pipeline and deploys.
Environment	Any modern browser, desktop or mobile. Build: Node.js ≥ 22.18, pnpm 10. Hosting: any static host.
Constraints	No paid services, no database server. Google Scholar has no public API and rate-limits scraping.
Assumptions	Public profile data may be shown. A data snapshot (not live data) is acceptable.
Interfaces	HTTP GET to iips.edu.in, scholar.google.com, api.openalex.org (pipeline only). The site itself makes no API calls.

4. Functional Requirements

ID	The system shall ...	Page
FR-1	Show portal totals (faculty, publications, citations, patents, Ph.D.), top scores, recent papers, top research areas	/
FR-2	List all faculty as cards	/faculty
FR-3	Search faculty by name, designation or research area; filter by department; sort by IIPS order, surname, score or citations	/faculty
FR-4	Show a profile: details, photo, links, Research Index, rank, percentile, radar vs IIPS median	/faculty/<slug>
FR-5	List a person's counted publications, filterable by type, plus the papers set aside with the reason	/faculty/<slug>
FR-6	Show citations per year, Ph.D. scholars, projects, memberships, responsibilities	/faculty/<slug>
FR-7	Rank all graded faculty; filter the ranking by department	/scores
FR-8	Explain the scoring: pillars, weights, curve, grades, what is not counted	/method
FR-9	Show every source-profile match with its evidence and confidence; filter to weak or missing ones	/data-check
FR-10	Fetch, match and clean data from IIPS, Google Scholar and OpenAlex into one JSON snapshot	pnpm data
FR-11	Let the maintainer pin / reject a profile and exclude a paper with a reason	scripts/data/*.json
FR-12	Give partial records (no research profile) no grade and say so	all pages

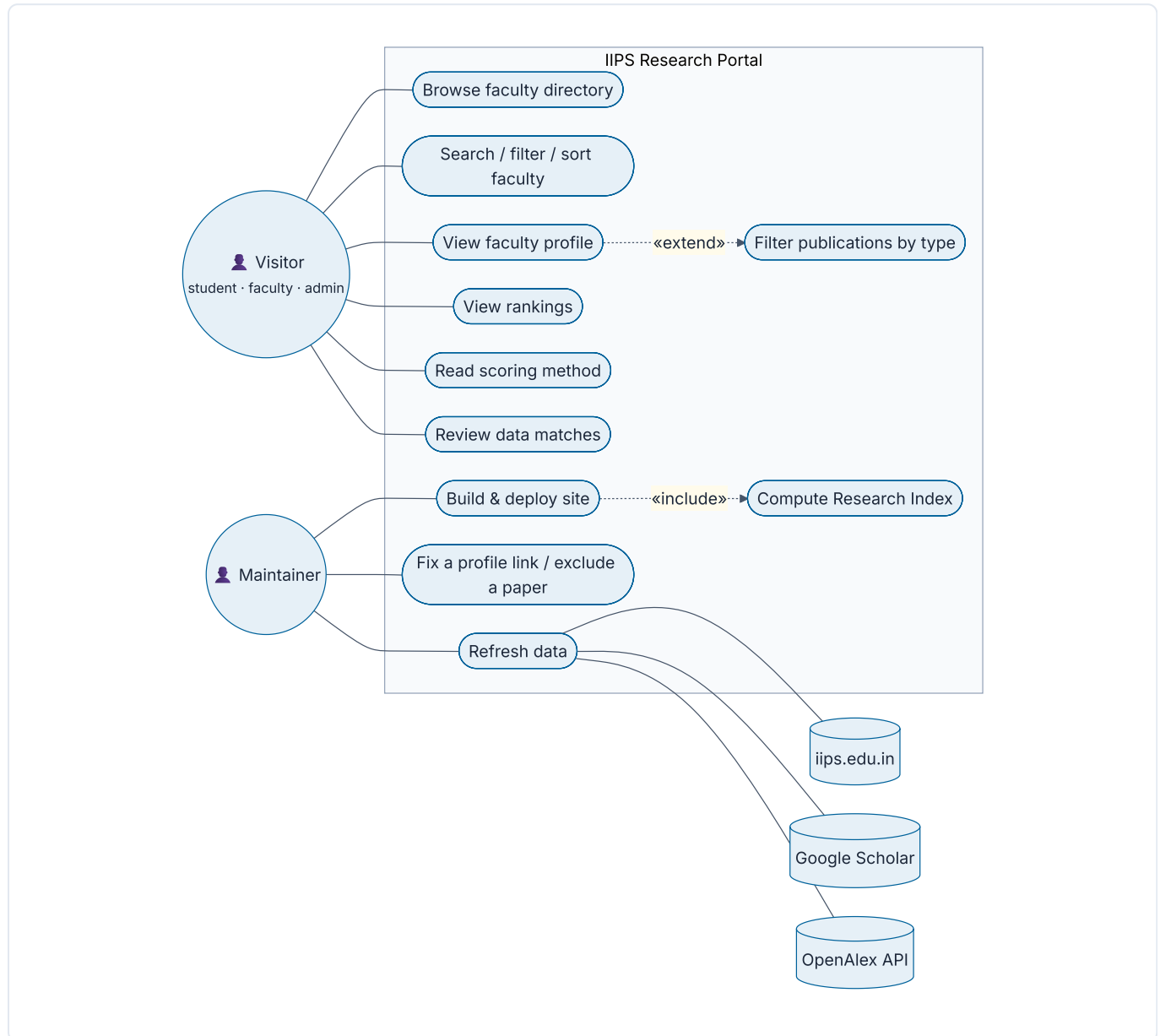
5. Non-functional Requirements

ID	Quality	Requirement and how it is met
NFR-1	Performance	Every page is prerendered HTML. No server work and no API calls when a page is opened.
NFR-2	Availability	Deployed to two hosts (GitHub Pages + Vercel) on every push to <code>main</code> .
NFR-3	Accessibility	WCAG AA colour contrast; automated axe checks on every page in the e2e tests.
NFR-4	Responsiveness	Works from 390 px phones to desktops (mobile e2e test).
NFR-5	Transparency	Score rules in one JSON file and on <code>/method</code> ; every match shows its evidence.
NFR-6	Data accuracy	Six cleaning rules drop namesake, duplicate and unconfirmed papers before scoring.
NFR-7	Maintainability	Strict TypeScript, ESLint + Prettier, domain logic kept apart from pages.
NFR-8	Testability	50 unit tests (scoring, cleaning, data) + 28 end-to-end tests, all passing.
NFR-9	Portability	Plain static files; runs on Windows, macOS, Linux; any static host.
NFR-10	Security & privacy	No backend, no logins, no secrets. Only public profile data is shown.
NFR-11	Cost	₹0: free hosting, free data sources.

6. UML Diagrams

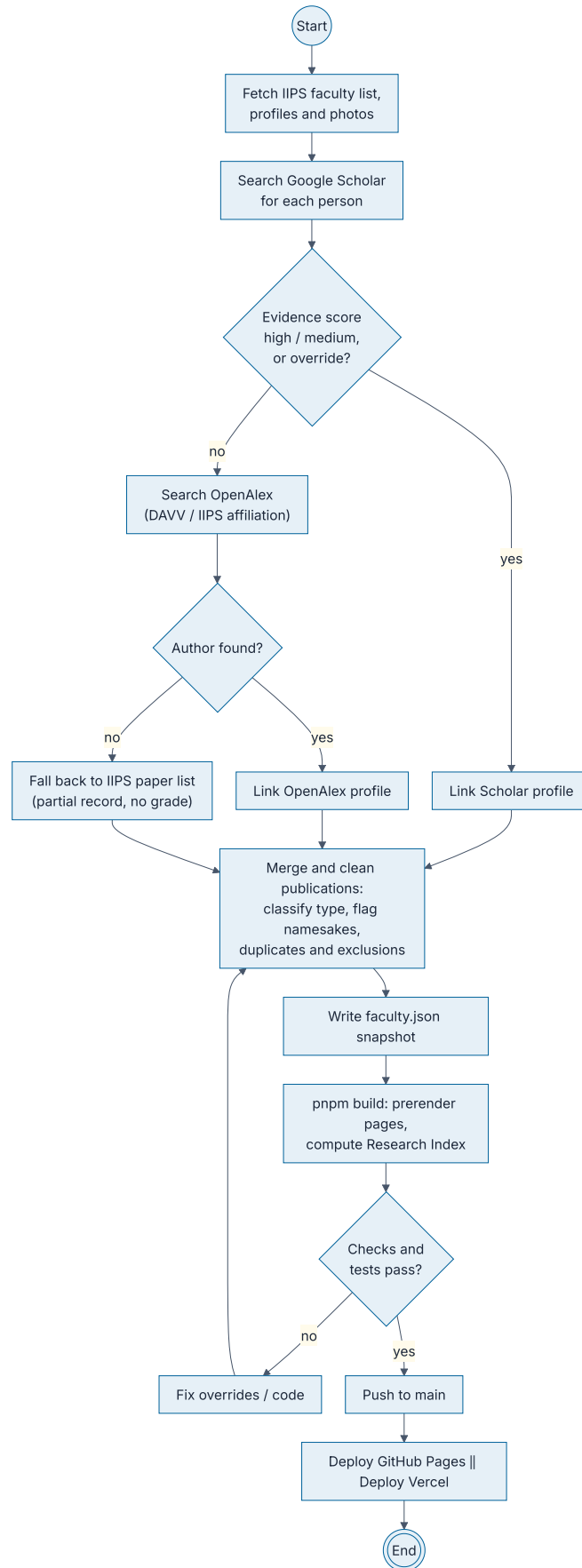
Drawn in Mermaid so they live next to the code and render on GitHub. Mermaid has no native use-case, object, communication, package, component or deployment notation, so those use flowchart shapes that follow the UML layout.

6.1 Use Case Diagram

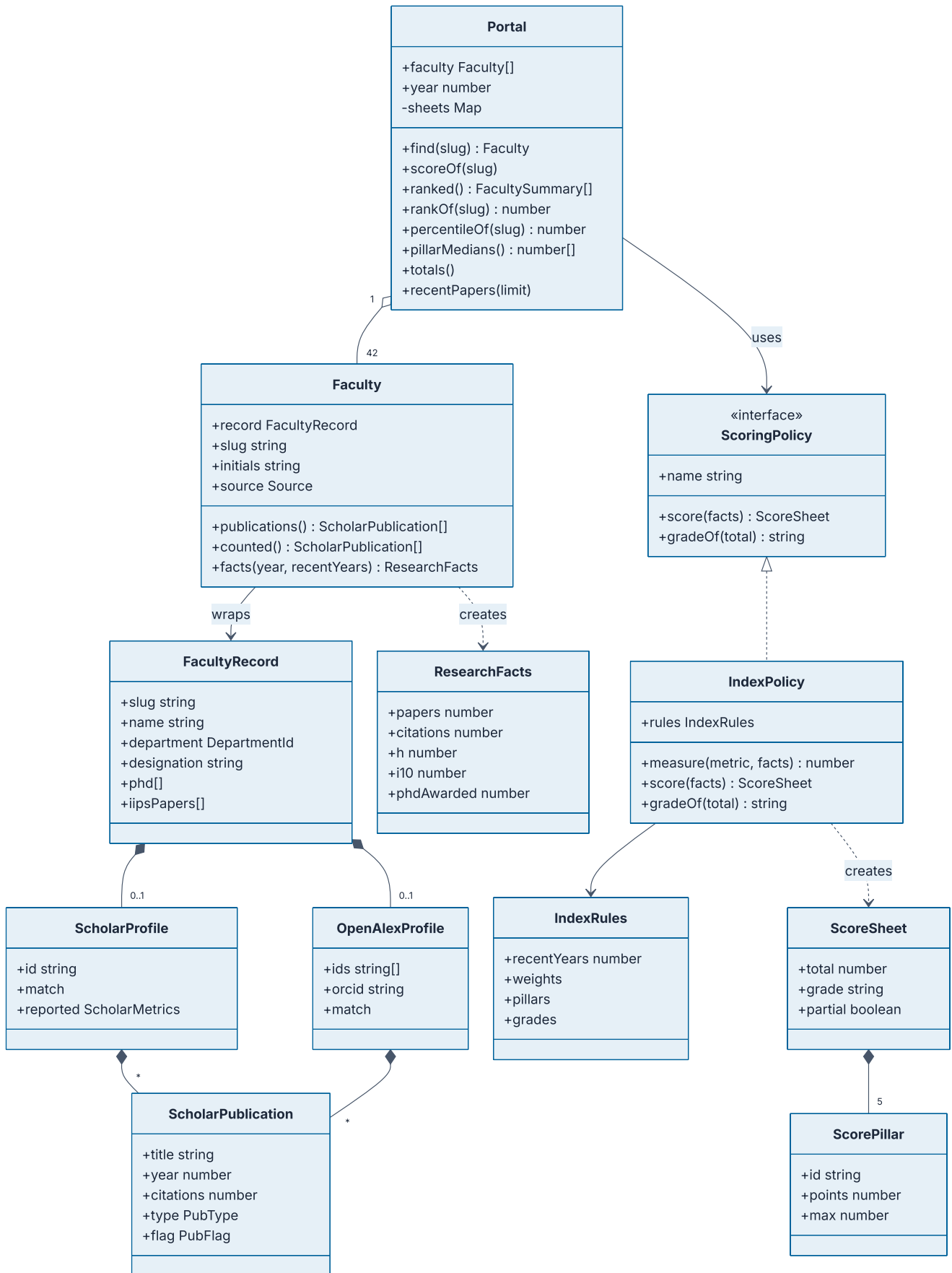


Actors: *Visitor* generalises student, faculty and admin (all read only). *Maintainer* runs the pipeline. The three data sources are secondary (system) actors.

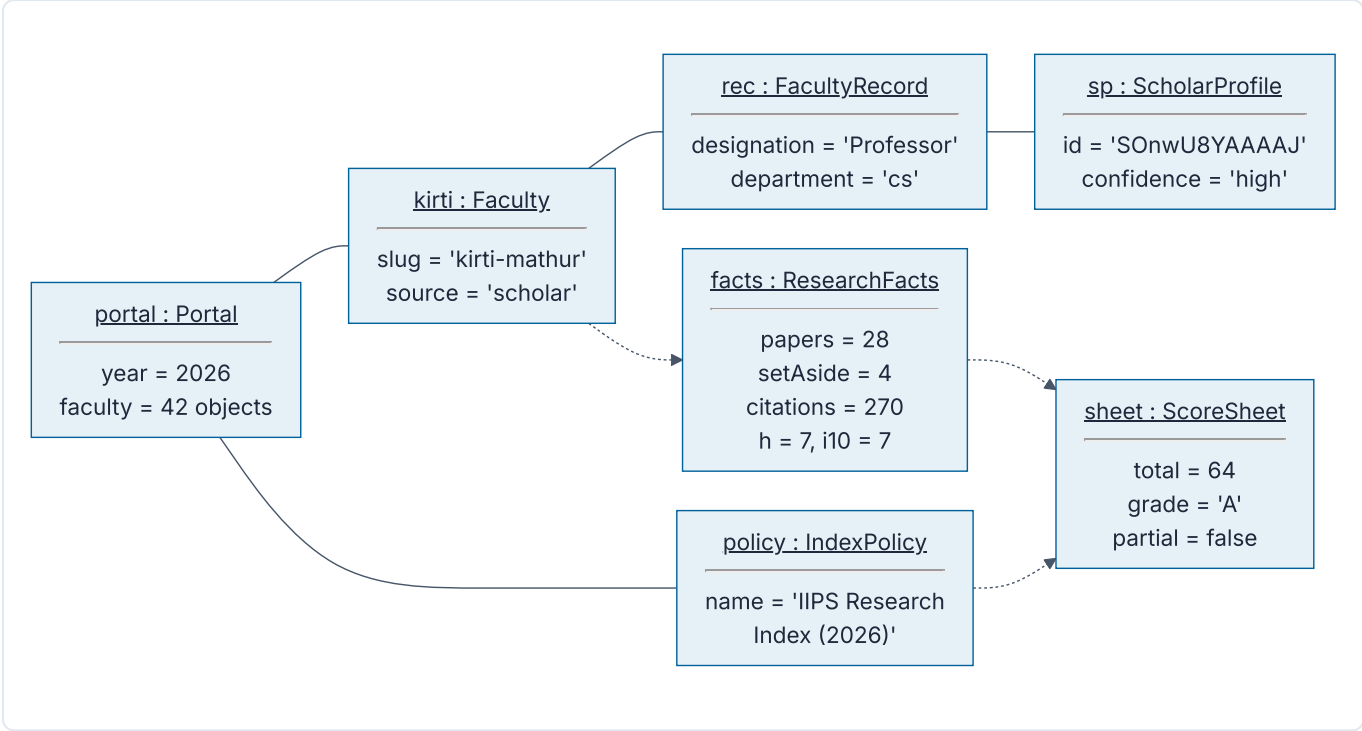
6.2 Activity Diagram — data refresh and publish



6.3 Class Diagram — domain and service layer



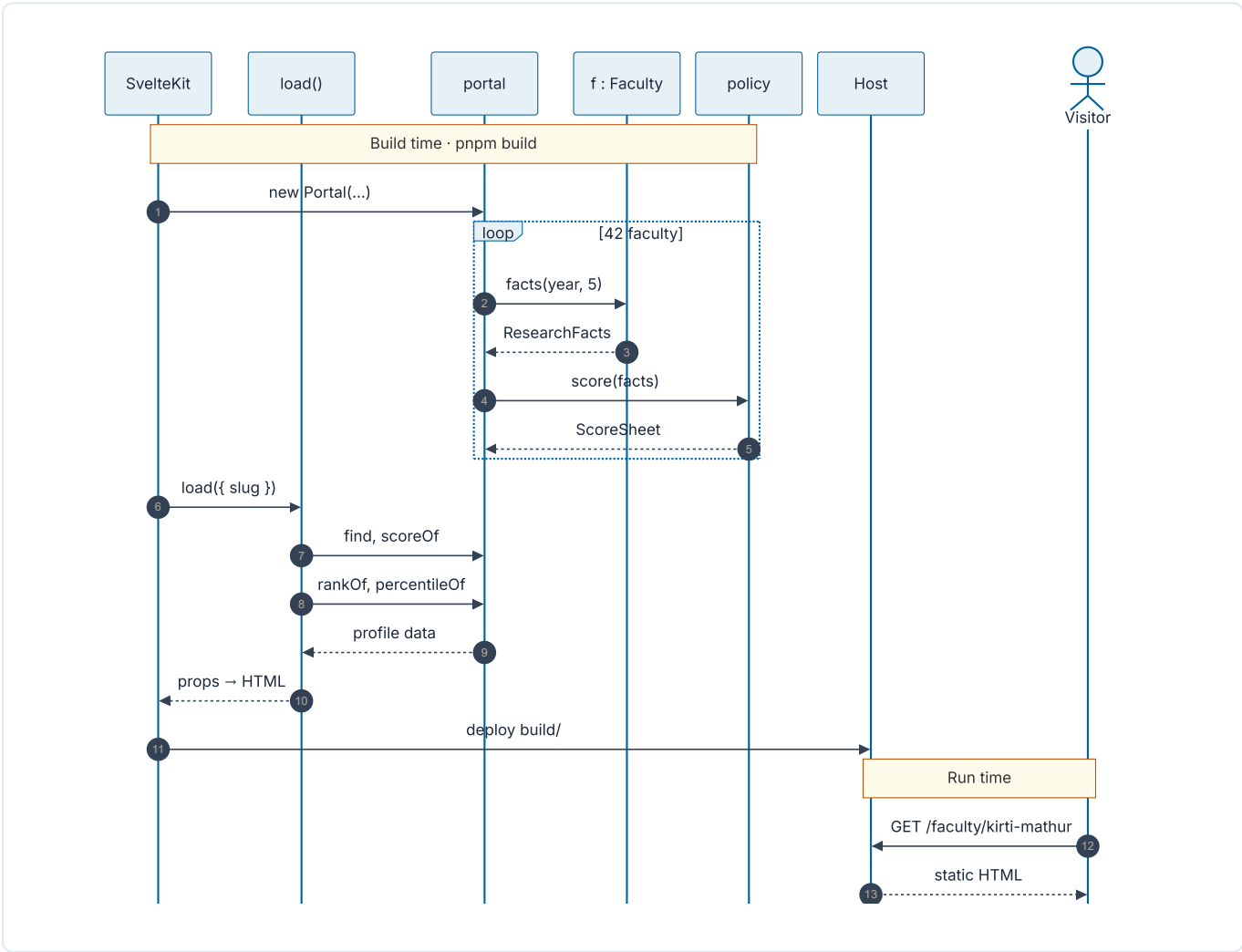
6.4 Object Diagram — one real profile at build time



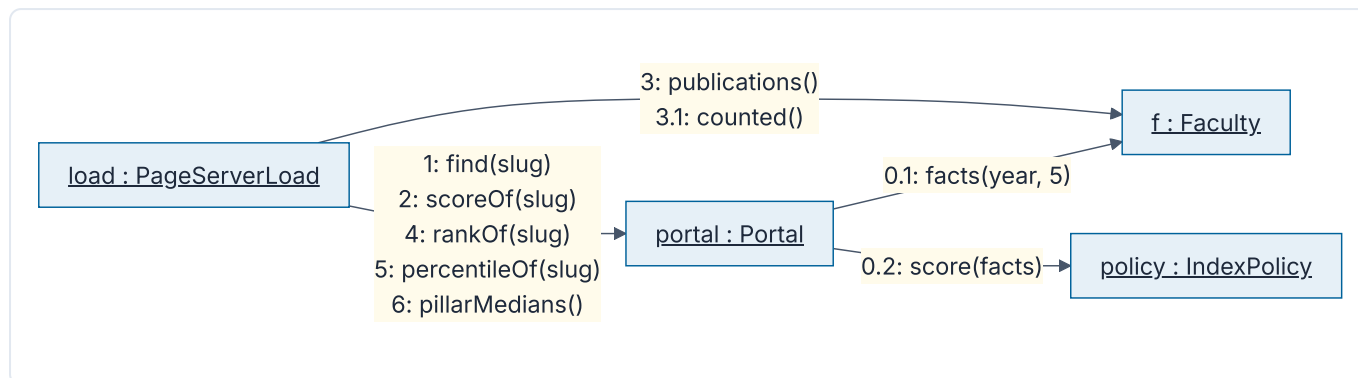
Values are real (snapshot of 2 Oct 2026): rank 3 of 26 graded faculty.

6.5 Sequence Diagram — building and viewing a profile

portal : Portal, policy : IndexPolicy, load() of /faculty/[slug] ; Host = GitHub Pages / Vercel.

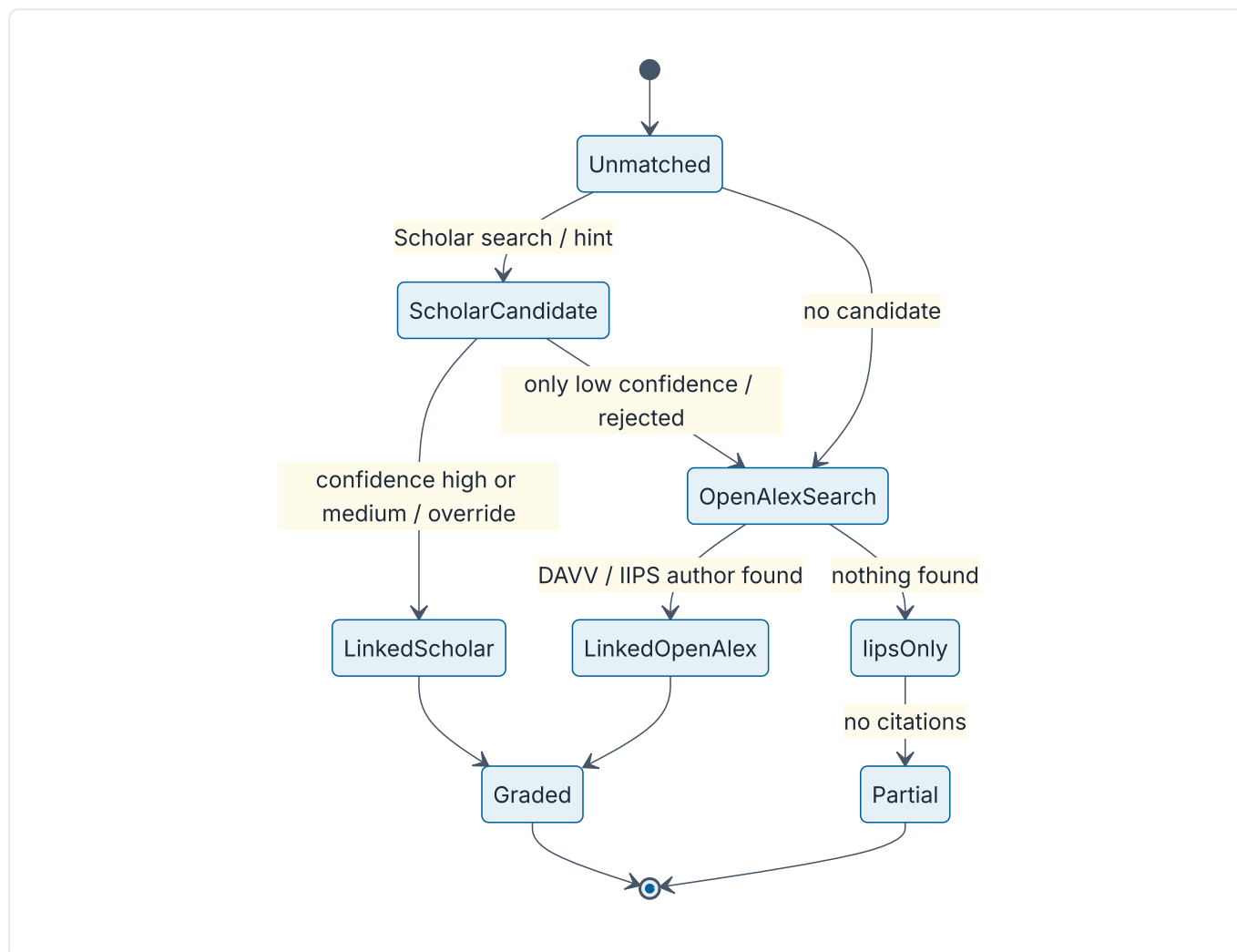


6.6 Communication Diagram — same interaction, object view



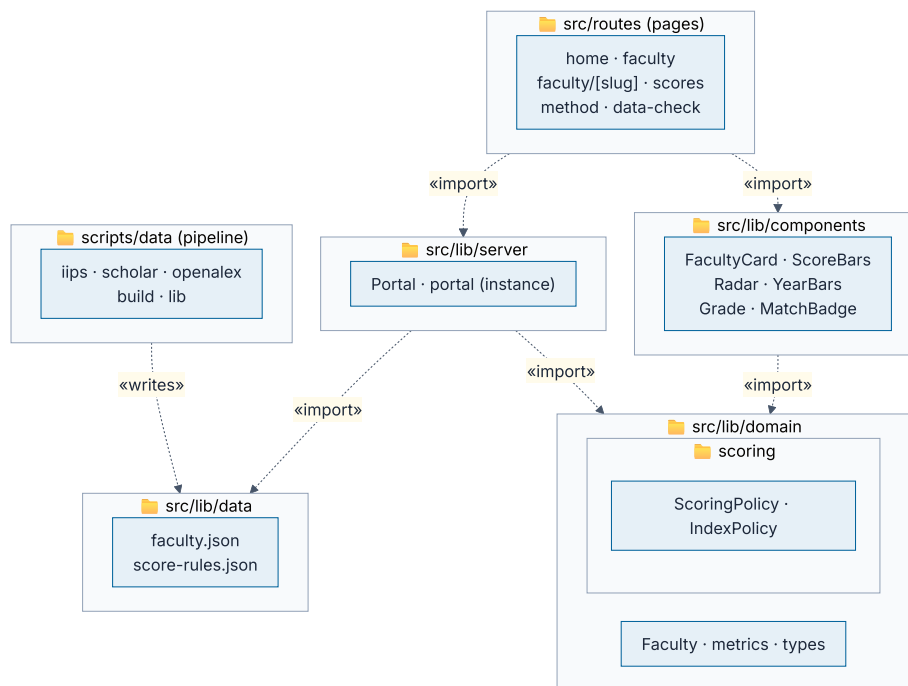
Messages `0.x` run once per faculty in the `Portal` constructor; `1-6` run for each profile page.

6.7 State Machine Diagram — a faculty member's data link

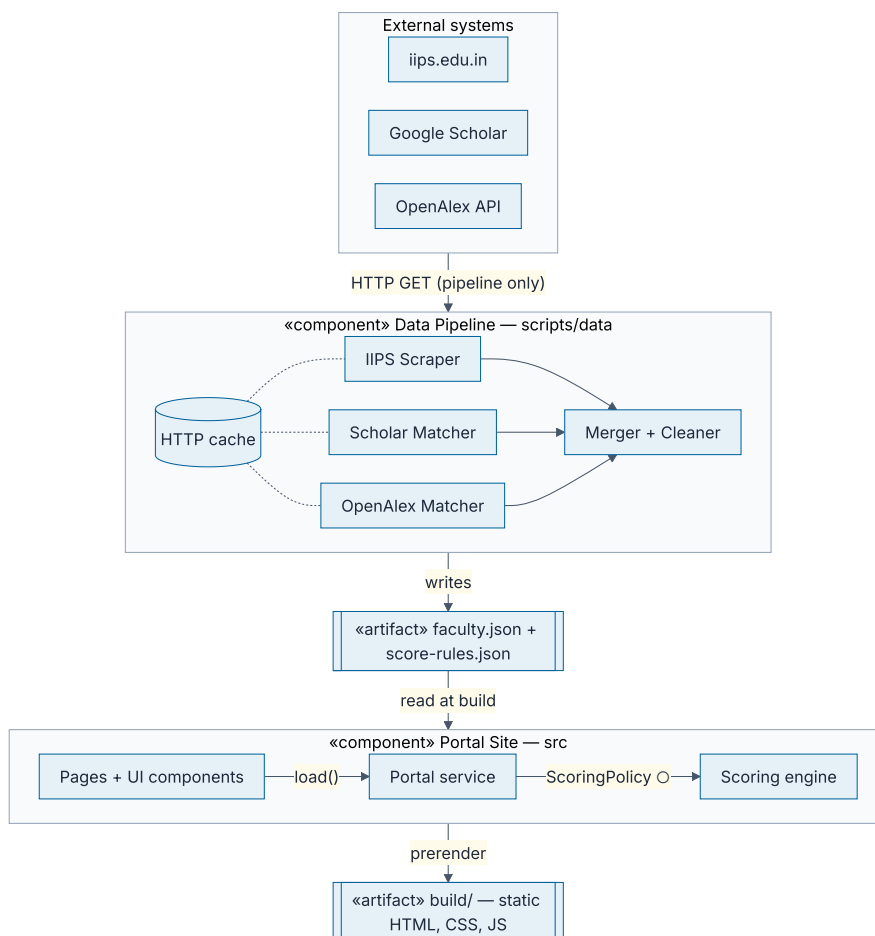


Snapshot: 19 linked to Scholar, 7 to OpenAlex, 16 IIPS-only (partial, no grade).

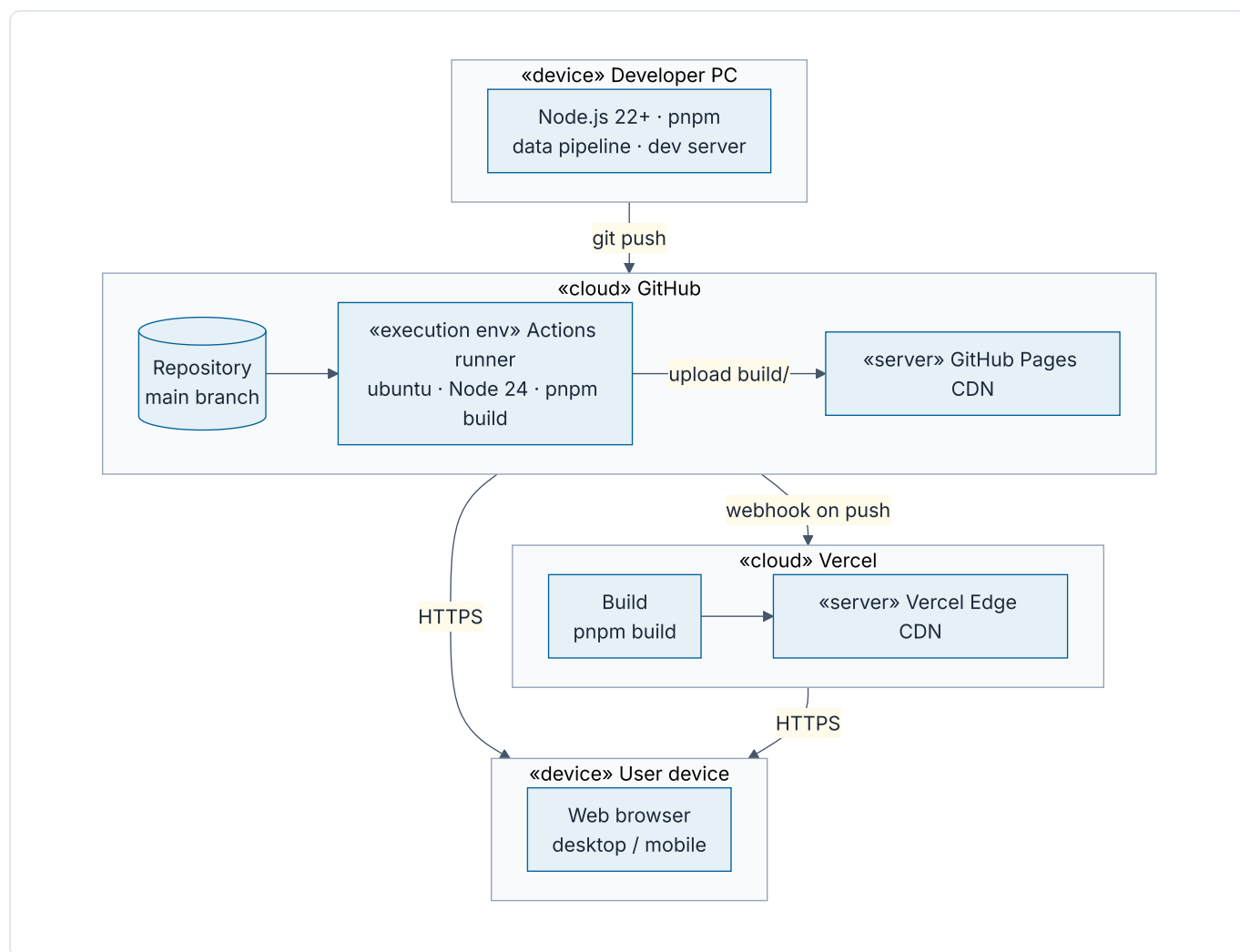
6.8 Package Diagram



6.9 Component Diagram

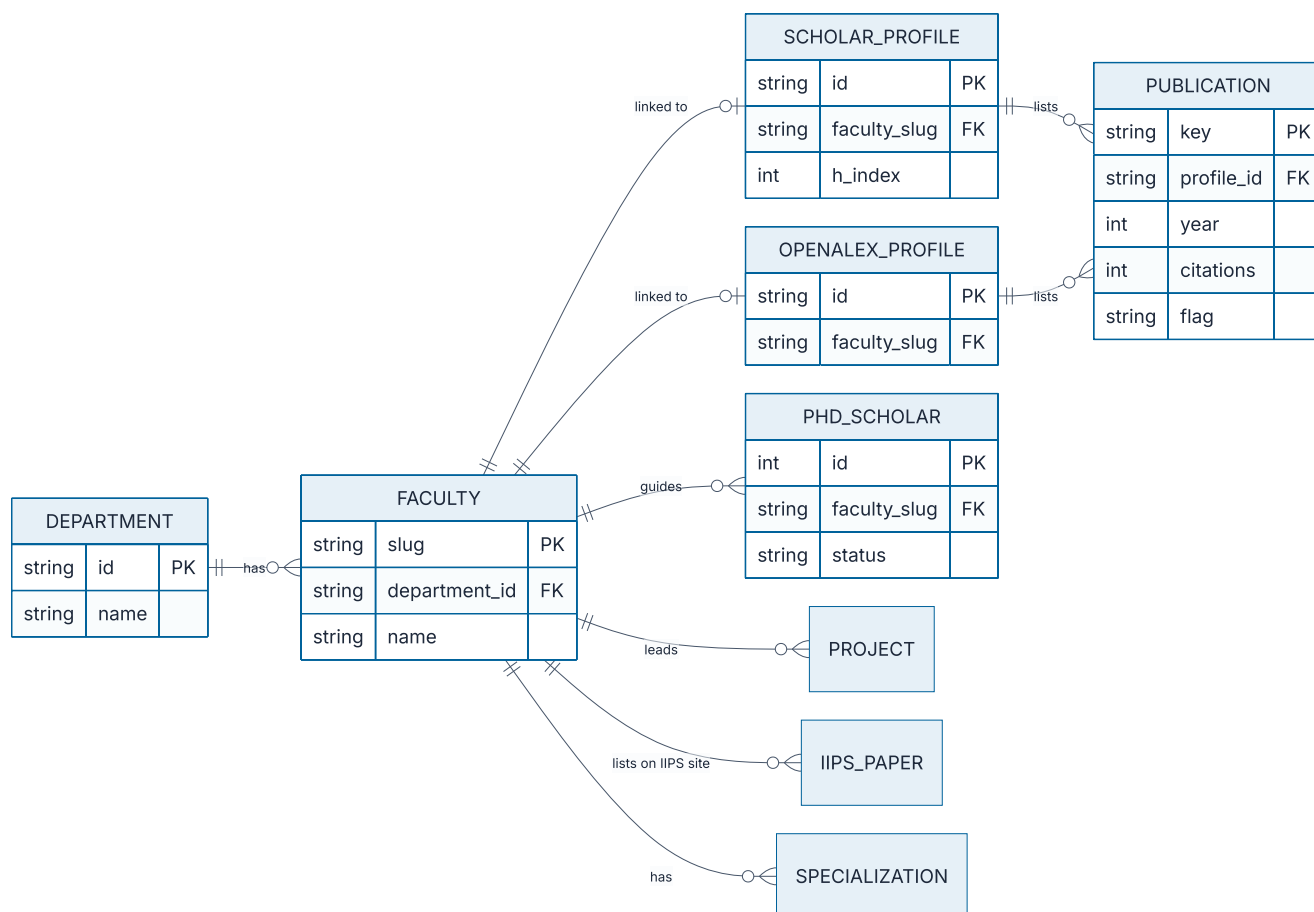


6.10 Deployment Diagram



7. Database Design

No database server is used, by design. The data is a single JSON snapshot read at build time (free hosting, nothing to run or secure). Below is the **logical** ER model of that JSON and the relational schema it would map to if moved to an RDBMS.



Relational schema

PK in **bold**, FK in *italics*. The ER diagram shows keys and main fields only; full columns are here.

Table	Columns
department	id , name, short
faculty	slug , iips_id, <i>department_id</i> , name, designation, qualification, teaching_years, photo, sort_order
specialization	<i>faculty_slug</i> , area
scholar_profile	id , <i>faculty_slug</i> (unique), confidence, decided_by, citations, h_index, i10_index
openalex_profile	id , <i>faculty_slug</i> (unique), orcid, confidence
publication	key , <i>profile_id</i> , source, title, authors, venue, year, citations, type, flag, doi
phd_scholar	id , <i>faculty_slug</i> , name, topic, status (awarded / submitted / ongoing)
project	id , <i>faculty_slug</i> , title, agency, status
iips_paper	id , <i>faculty_slug</i> , text, link

Score rules are configuration (`score-rules.json`), not data, so they are kept out of the schema.

8. UI Mockups / Wireframes

Inspiration: the visual style follows the **IIPS-CoMET 2027** conference site, iips.edu.in/comet27: navy and IEEE blue, amber call-to-action buttons, small uppercase labels above headings, IIPS + DAVV logos in the header. Amber was darkened so white text on it passes WCAG AA.

Separate low-fidelity wireframes were **not drawn** (time). The layout was built straight in code from the CoMET style, so the screenshots of the working prototype below serve as the high-fidelity mockups.

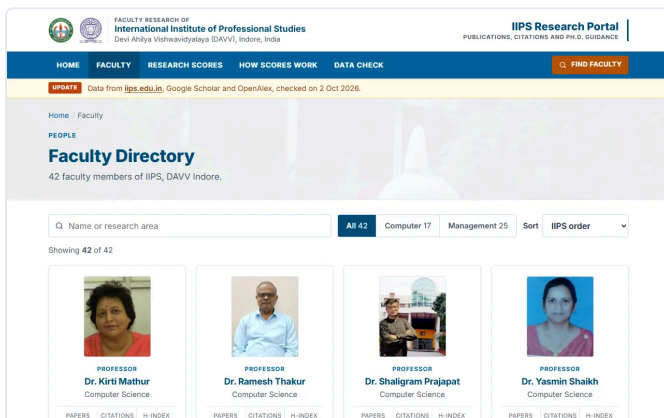
Home



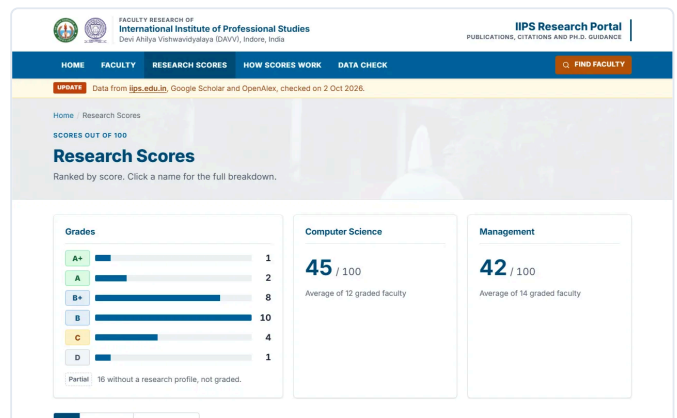
Faculty profile



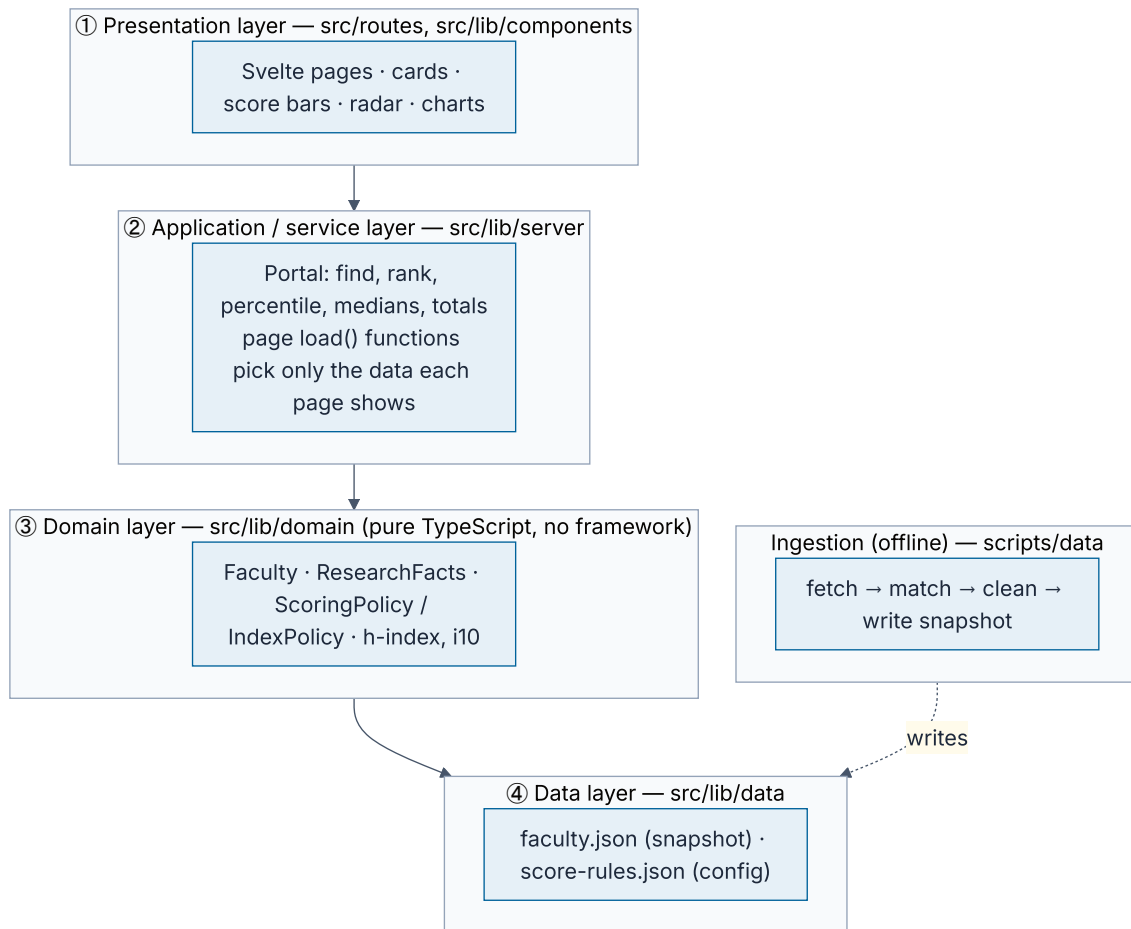
Faculty directory



Research score rankings



9. Layered Architecture



Rule: each layer only calls the one below. The domain layer has no imports from SvelteKit, so scoring is unit-tested on its own.

10. Design Patterns

Pattern	Where	Why it helps
Strategy	<code>ScoringPolicy</code> interface ← <code>IndexPolicy</code>	Swap or add a scoring rule set (e.g. an official API score) without touching pages.
Repository + Facade	<code>Portal</code>	Pages ask one object for ranks, scores and papers; JSON details stay hidden.
Singleton	<code>portal</code> instance in <code>src/lib/server/index.ts</code>	Snapshot is loaded and scored once per build, shared by every page.
Adapter / wrapper	<code>Faculty</code> wraps <code>FacultyRecord</code>	Three sources (Scholar, OpenAlex, IIPS) look the same to the rest of the code via <code>source</code> , <code>publications()</code> , <code>facts()</code> .
Pipes and Filters	<code>scripts/data</code> : <code>iips</code> → <code>scholar</code> → <code>openalex</code> → <code>build</code>	Each step reads the previous output file; any step can be rerun alone.
Cache-Aside	<code>cachedFetch()</code> in <code>scripts/data/lib.ts</code>	Never hits a site twice; survives Scholar rate limits.
Data-driven configuration	<code>score-rules.json</code>	Weights, targets and grades change without code changes.

11. Software Design Document (SDD)

Area	Design decision
Architecture	Static site generation: all pages prerendered from one snapshot (§9). No runtime server.
Data flow	Pipeline (§6.2) → <code>faculty.json</code> → <code>Portal</code> → pages → static HTML (§6.9).
Profile matching	Scholar candidates get evidence points (name fit, IIPS / DAVV affiliation, email domain, shared papers, Ph.D. co-authors) → high / medium / low. Manual <code>hints</code> and <code>overrides</code> win.
Cleaning rules	A paper is set aside if: before-career, not-author, duplicate, manual, other-field, unconfirmed.
Research Index	5 pillars × 20 = 100: Output, Impact, Quality, Recent work (5 yrs), Mentoring .
Score curve	<code>points = max × min(1, √(value ÷ target))</code> . Reaching ¼ of the target already gives ½ the points.
Paper weights	journal 1 · book 2 · patent 2 · conference / chapter / other 0.5 · thesis 0. Ongoing Ph.D. = 0.5 of awarded.
Grades	A+ ≥ 75 · A ≥ 60 · B+ ≥ 45 · B ≥ 30 · C ≥ 15 · D. IIPS-only records: score shown, no grade .
Ranking	Graded first, by total, then citations, then IIPS seniority order.
Error handling	Unknown slug → 404 page; missing source → partial record with a visible note.
Testing	Vitest unit tests for scoring, cleaning, data integrity; Playwright e2e + axe accessibility.
Deployment	GitHub Actions → GitHub Pages, and Vercel; both on every push to <code>main</code> (§6.10).

12. Prototype, GitHub & Documentation

Item	Link / command
Live prototype	arpanpatra111.github.io/IIPS-Research · iips-research.vercel.app
Repository	github.com/ARPANPATRA111/IIPS-Research
Run locally	<code>pnpm install</code> → <code>pnpm dev</code> → http://localhost:5173
Quality checks	<code>pnpm check</code> · <code>pnpm lint</code> · <code>pnpm test</code> · <code>pnpm test:e2e</code>
Refresh data	<code>pnpm data</code> (needs network) or <code>pnpm data:build</code> (offline, from saved fetches)
Documentation	<code>README.md</code> (setup, deploy, data, layout, limits) + this report + the <code>/method</code> page

13. Not Done, and Why

Item	Status	Reason
Login, roles, faculty self-editing, admin CRUD	Not built	Needs a backend and database; out of time. Corrections go through <code>overrides</code> / <code>exclusions</code> JSON + rebuild.
Live / real-time data	Not possible	Google Scholar has no public API and blocks frequent automated requests. A dated snapshot is used.
Official UGC API score / PBAS	Replaced	Official scoring needs self-appraisal forms and proof documents that are not public. A transparent local index is used instead.
Patent tracking	Partial	Only 2 patents found: Scholar / OpenAlex rarely list patents and there is no public IIPS patent feed.
Citations for 16 faculty	Partial	No Scholar or OpenAlex profile could be confirmed; shown as partial, never graded against others.
Relational database	Logical only	Not needed for a read-only static site; ER + schema given in §7.
Low-fidelity wireframes	Skipped	Time; layout built directly from the CoMET reference, screenshots used as mockups (§8).